

Education

Tufts University

Bachelor of Science, Mechanical Engineering GPA: 3.3

Medford, MA

May 2027

Engineering Experience

Nolop FAST Facility

Makerspace Engineering & Development Intern

Medford, MA

Summer 2026

- Built a Raspberry Pi/Flask particulate-monitoring system using a Dylos DC1100 Pro to log and visualize woodshop air quality against OSHA/NIOSH guidance.
- Planned and conducted before-and-after testing of individual woodshop equipment and multi-day facility use to evaluate the effectiveness of upgraded dust collection and air-filtration systems
- Supported 200+ pre-college students through CAD, 3D printing, laser cutting, woodshop instruction, design reviews, safety training, and facility operations.

Fabrication Specialist

Jan 2026 - Present

- Advise students and researchers on material selection, tolerance stack-ups, DFM, and fabrication using 3D printing, laser cutting, and CNC milling.
- Provide hands-on equipment training and enforce safety protocols in a high-throughput makerspace.

Derby Entrepreneurship Center, Tufts University

Teaching Assistant, ENT164: Intro to Making

Medford, MA

Aug 2024 - Present

- Guide 50+ students per semester through full design-build-test cycles of electromechanical systems
- Troubleshoot integration issues involving sensors, actuators, and microcontrollers under real project constraints
- Iterated course materials and microcontroller transitions across 4+ semesters to improve clarity and build success

Tufts Center for Engineering Education and Outreach (CEEEO)

Teaching Fellow, Engineering Design Lab

Medford, MA

Summer 2025

- Led high school students in building robotic systems using Python, Raspberry Pi, and digital fabrication
- Designed and taught IoT and control system modules using Jupyter Notebooks
- Built demos, refined curriculum, and mentored project teams to develop fully integrated mechatronic systems

Leadership & Activities

Zeta Beta Tau Fraternity, Chapter President

Nov 2025 - Present

Tufts x MIT Leadership Training Institute, Head Leadership Mentor

Jan 2025 - Present

Fitness & Individual Development at Tufts (F.I.T.) Pre-Orientation, Peer Leader

Aug 2024 - Present

Projects

Autonomous Shop Vac, Nolop FAST Facility

Led drivetrain design for an autonomous Ryobi shop vacuum, developing motor mounts and a belt-tensioning system; integrated Raspberry Pi controls, motor electronics, and LiDAR-based navigation for automated dust collection in the Nolop Fast Facility's woodshop

Assistive Design

Designed a low-force positioning/actuation aid enabling a client with cerebral palsy to independently apply hair detangler

Skills

Fabrication & Manufacturing: 3D Printing, Laser Cutting, CNC Machining, Rapid Prototyping

Hardware & Embedded Systems: Raspberry Pi, ESP32, GPIO interfacing, common sensors & actuators

Programming: Python (MicroPython + CircuitPython), Jupyter Notebooks, MATLAB, LabVIEW

Software: CAD (Solidworks, Onshape, AutoCAD), Solidworks FEA, KiCAD, COMSOL Multiphysics, 3DS Max

Interests: Tufts Climbing Team member, Photography, teaching/mentorship, solving Rubik's Cubes, Open-format DJing